

FILTER DESIGN TEST CASE

Low-Pass Filter - Band 12 (700 MHz) for Public Safety Band Filter
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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1. OBJECTIVE

This document describes the design, simulation, and characterization of a Low-Pass Filter filter targeting Band 12 (700 MHz) for Public Safety Band Filter. The filter is designed on 42° YX LiTaO3 substrate with a target center frequency of 1758.5 MHz and bandwidth of 293.9 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the Public Safety Band Filter module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Low-Pass Filter
- Frequency Band: Band 12 (700 MHz)
- Application: Public Safety Band Filter
- Substrate: 42° YX LiTaO3
- Package: CSP (Chip Scale Package)
- Design Tool: Keysight ADS 2024
- Design Center: Andover HQ Lab

This specification applies to all variants of the Low-Pass Filter design targeting Band 12 (700 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	1758.5 MHz
Bandwidth (3 dB):	293.9 MHz
Insertion Loss Target:	<= 3.2 dB
Return Loss Target:	>= 23.0 dB
Out-of-Band Rejection:	>= 30.0 dB
Isolation (Duplexer):	>= 34.0 dB
Q Factor:	3152
Temperature Coefficient:	-27.3 ppm/°C
Die Size:	2.4 x 1.0 mm
Wafer Size:	6-inch
Number of Resonators:	10
Electrode Material:	Al

Passivation: SiO2

4. TEST PROCEDURE

1. Open Keysight ADS 2024 and load the Low-Pass Filter template project.
2. Define the substrate stack: 42° YX LiTaO3 with specified layer thicknesses.
3. Set design targets: center freq 1758.5 MHz, BW 293.9 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 3.2 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (CSP (Chip Scale Package)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (6-inch, 42° YX LiTaO3).
10. Perform on-wafer probing using Anritsu MS46122B ShockLine.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 5276 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 3.2 dB
- Return loss in passband shall be >= 23.0 dB
- Out-of-band rejection at +/- 441 MHz offset >= 30.0 dB
- Group delay variation across passband shall be <= 4 ns
- Temperature drift over -40°C to +85°C shall be <= 6.0 MHz
- Power handling shall be >= +29 dBm at 1 dB compression
- ESD tolerance >= 500 V (HBM)
- Die yield on qualification lot >= 92%

6. TEST RESULTS

Measurement Date: 2025-06-06
Devices Tested: 91
Insertion Loss (meas): 2.94 dB
Return Loss (meas): 24.6 dB
Rejection (meas): 32.5 dB
Isolation (meas): 39.0 dB
Q Factor (meas): 3152
Group Delay Variation: 2.9 ns
Power Handling: +31 dBm
Die Yield: 94.5%

Result Classification: FAIL

7. OBSERVATIONS AND FINDINGS

- a) The Low-Pass Filter design demonstrated below-target performance for Band 12 (700 MHz).
- b) Insertion loss is marginally above target; electrode thickness optimization recommended.
- c) Out-of-band rejection exceeded the minimum requirement.
- d) Slight frequency drift observed at +85°C; TC layer thickness may need adjustment.
- e) Cross-talk between adjacent filter channels was below -55 dB.

8. CORRECTIVE ACTIONS

- 1. Optimize resonator geometry to reduce insertion loss by ~0.3 dB.
- 2. Re-run EM simulation with refined mesh for better accuracy.
- 3. Escalate to advanced materials team for alternative piezoelectric stack.

9. SIGN-OFF

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Approver:	Dr. James Liu	Date:	2025-06-19

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